Seungwoo Han

M.S., Naval Architecture and Ocean Engineering, Seoul National University
nimp91@snu.ac.kr — Website — Google Scholar — GitHub

Research Interests

Research Focus: Simulation-based modeling · Complex engineering systems · System optimization
Methods: Sequential decision-making · Reinforcement learning · Graph neural networks
Applications: Maritime autonomous systems · Industrial logistics · Shipyard systems

Education

Seoul National University Seoul, Republic of Korea
M.S. in Naval Architecture and Ocean Engineering 2025
GPA: 3.9 / 4.3
Thesis: Graph Reinforcement Learning for Integrated Optimization of Shipyard Hull-Block Logistics

Seoul National University Seoul, Republic of Korea
B.S. in Naval Architecture and Ocean Engineering, Minor in Industrial Engineering, 2023
GPA: 3.49 / 4.3

Publications

Journal Publications

Han, S. W., Oh, S. H., and Woo, J. H. "Crystal Graph Convolutional Neural Network-Based Reinforcement Learning for Adaptive Shipyard Block Transportation Scheduling." *Engineering Applications of Artificial Intelligence*, 2026.

Oh, S. H., Cho, Y. I., Oh, H., Baek, J., **Han, S. W.**, and Woo, J. H. "Framework for State Features Design in Job Shop Scheduling with Deep Reinforcement Learning: Beyond Empirical Approaches." *Journal of Computational Design and Engineering*, 2026.

Han, S. W., Kwak, D. H., Byeon, G. W., and Woo, J. H. "Forecasting Shipbuilding Demand Using Shipping Market Modeling: A Case Study of LNGC." *International Journal of Naval Architecture and Ocean Engineering*, 2024.

Manuscripts Under Review

Han, S. W., Han, Y., Yoo, Y., Yoon, H., and Woo, J. H. "Optimization of Hull-Block Placement and Retrieval in a Congested Planar Storage System." Under review at *Computers & Industrial Engineering*, 2026. Preprint available at SSRN.

Manuscripts in Preparation

Han, S. W., Yeo, H. B., and Oh, S. H. "Current-Aware Rendezvous Planning for Multiple Patrolling USVs under Heading-Constrained Maritime Navigation." Manuscript in preparation for submission to ICRA 2027.

Research Experience

Research Assistant / Mandatory Military Service 2025–Present
Naval Future Innovation Research Center, Republic of Korea Navy Republic of Korea

- Formulated a priority-aware rendezvous optimization problem for multiple patrolling unmanned surface vehicles and constructed a simulation-based maritime environment.
- Built a grid-based maritime navigation graph with edge costs reflecting ocean currents and vessel turning constraints, and evaluated Dijkstra and A* search-based path-planning baselines.
- Developed graph neural network and reinforcement learning methods for priority-aware rendezvous sequence optimization, using path-planning-derived travel times as transition costs.

Graduate Researcher 2023–2025
Seoul National University Seoul, Republic of Korea

- Formulated shipyard hull-block transportation and storage/retrieval operations as two interconnected routing and scheduling optimization problems.
- Developed graph neural network-based reinforcement learning approaches to solve the transportation scheduling and placement/retrieval planning problems.
- Built an integrated simulation framework for joint optimization of hull-block transportation, storage, and retrieval decisions in shipyard logistics.
- Developed a structured maritime market model for shipbuilding demand forecasting by integrating shipping market domain knowledge, market indicators, and historical data.

Technical Skills

Programming and Simulation: Python, C++, PyTorch, Unity, NumPy, Pandas

Modeling: Mixed-Integer Programming, Markov Decision Processes, simulation-based modeling, discrete-event simulation, system dynamics

Learning-Based Methods: Reinforcement learning, policy gradient and actor-critic methods, graph neural networks, transformers, decision tree models

Search and Optimization Methods: Graph search, dynamic programming, greedy heuristics, metaheuristic optimization

Applications: Shipyard logistics, maritime autonomous systems, multi-USV coordination, industrial systems optimization

Honors and Scholarships

American Bureau of Shipping Scholarship	2024
Brain Korea 21 Outstanding Graduate Student Scholarship	2024
Merit-Based Scholarship, Seoul National University	2022, 2023, 2025
Kwanak Foundation Scholarship, Seoul National University Alumni Association	2021